

*True third-generation
upgraded version*



欣易云联
Sinilink

XY7025

DC **buck** power supply **1750W**

CC • CV • CW

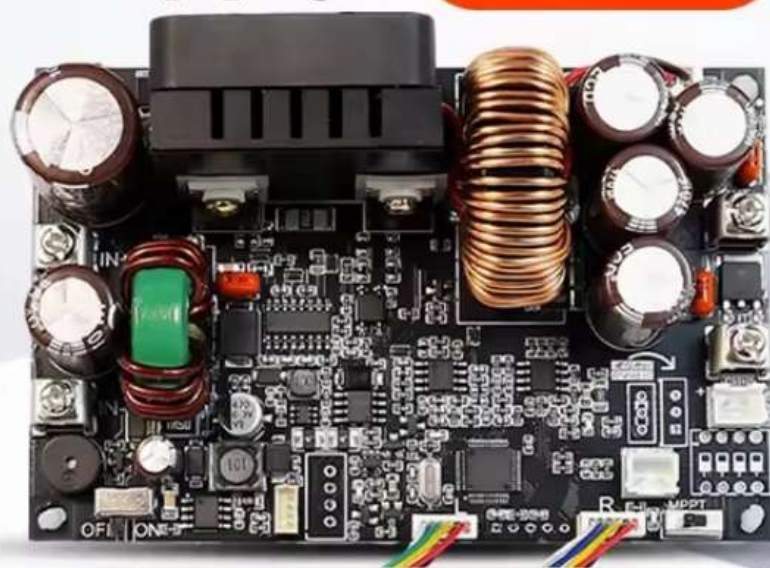
Input voltage **12-85.00v**

Output voltage **0-70.00v**

Output current **0-25.00A**

Power output **1750w**

Storage space **10 groups**



**simply the all-viewing
-angle VA color LCD**

- Support firmware upgrade
- support MPPT solar charging

The LCD screen is more than 1.8 inches!

Supports the standard Modbus -protocol for serial communication
Independent dual serial ports, serial port TTL and on-board 485 communication

XY7025 Bare board

The bare board can be used independently

Output voltage 0-70V Output current 0-22A

Independent dual serial ports

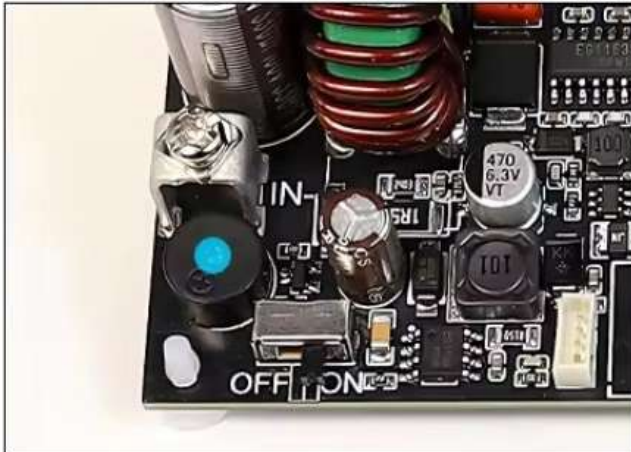
TL level communication and on-board 485 communication



NOTE: When purchasing a bare device, switches, potentiometers, and dial switches will be welded on. If a set with screen meter is purchased, they are not welded by default.

Product details

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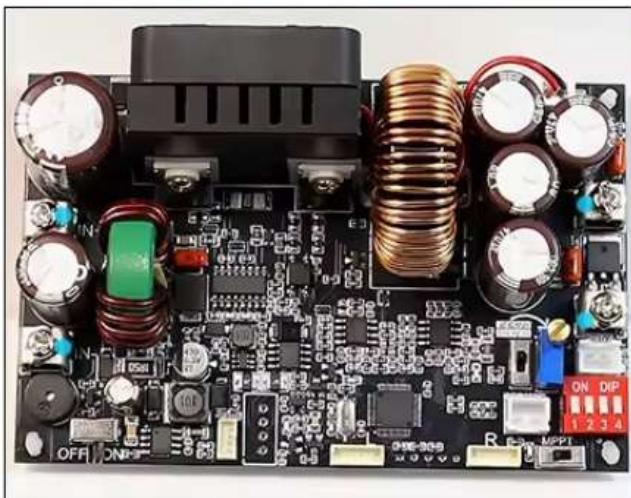
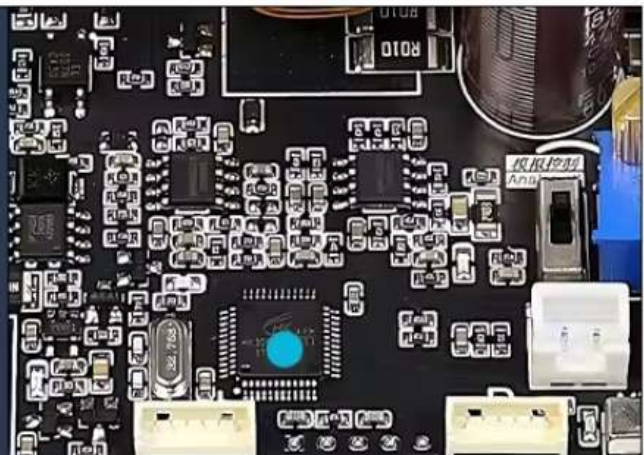


Onboard buzzer

Key prompt alarm prompt

High-precision external clock crystal oscillator.

Ensuring time accuracy and more precise capacity/energy statistics. Powerful 48-pin MCU controller with 64M frequency for faster calculation speed.



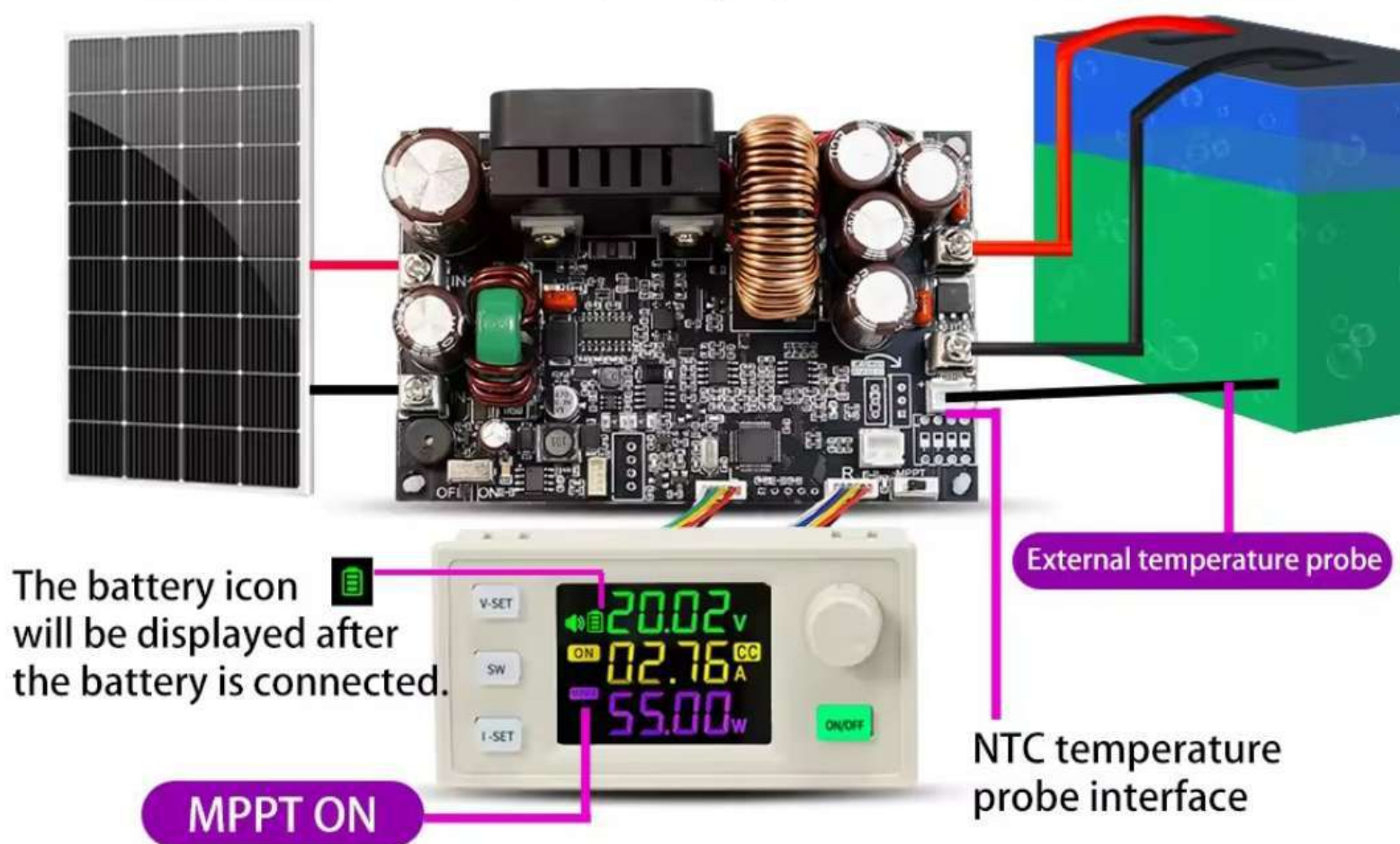
Upgraded PCB Welded terminals (M3)

More current
less heat
more reliable

Without anti-backflow function, the output terminal can be connected with an anti-backflow diode to charge various rechargeable batteries (if you are a professional, you can connect it without the anti-backflow diode).

With MPPT function, it supports MPPT solar charging

External temperature probe can be connected, supporting over-temperature protection. Attach the external temperature probe to the battery, and it will automatically stop charging in case of over-temperature.



Attention: This product does not have output reverse connection protection. Reversing the positive and negative terminals of the battery will damage the device. Battery charging requires certain professional knowledge. Non-professionals are not allowed to charge directly to prevent fire and explosion.

Product parameters

Product name	DC buck- power supply	Model	XY7025
Input voltage	12.00~85.00V	Output voltage	0~70.00V
Output current	0~25.00A	Accuracy of voltage	±0.5%+1 word
Power output	1750W	Accuracy of current	±0.5%+3 word
Resolution of voltage	0.01V	Resolution of current	00.01A
Storage data group	10 groups	Soft start	YES
Screen size	Over 1.8-inch LCD with a viewable area of 38*29mm		
Output ripple typical value	vpp-150mv		
MPPT function	Support MPPT solar charging	Buzzer	YES
Efficiency of conversion	About 95%	Number of buttons	5
Bare board size	111*72*45mm	Bare board weight	203g
Screen instrument	86*45*50mm	with packaging	230g
Bare board package	142*107*46mm	Product weight	250g
product Package	172*95*60mm	with packaging	310g

Protection mechanism

under-voltage(LVP)	Adjustable 10-95V, default 10V
over-voltage(OVP)	Adjustable from 0 to 72V, default is 72V
over-current(OCP)	Adjustable from 0 to 27A, default 27A
over-power(OPP)	Adjustable from 0 to 2000W, default is 1800W
over-temp(OTP)	Adjustable from 0 to 110 °C , default is 95 °C
over-time out(OHP)	1 minute - 99 hours 59 minutes, off by default
over-capacity(OAH)	Adjustable from 0 to 9999Ah, disabled by default
over-energy(OPH)	Adjustable from 0 to 4200KWh, off by default

Precautions:

1. The input cannot be connected to AC power, **nor can it be connected to 220V mains power.**
2. The maximum input voltage must not exceed 90V. It is recommended to use within 85V, and the positive and negative terminals must not be connected reversely.
3. The output has no reverse current protection. When connecting the battery, ensure that the power supply's output voltage is always higher than the battery voltage. Alternatively, a diode can be connected in series with the positive output terminal to allow one-way conduction, preventing reverse current from the battery voltage from damaging the power supply.
4. When connecting the battery to the output terminal, pay attention not to connect the positive and negative terminals reversely.
5. This product is a **DC buck power supply**. The maximum output voltage = input voltage * 0.945.

Top-quality materials and first-class texture

The wiring terminals are upgraded to PCB welded terminals.
More current, less heat, more reliable

Input common-mode inductors significantly reduce the impact of input power ripple, achieving low ripple output.

intelligent Temperature-Controlled Fan with Large-area ventilation heat sinks

Employ high-power, high-temperature-resistant iron-silicon-aluminum magnetic ring inductors.

Output 2-stage filtering

OUT+

OUT-

Support 0.3-10V analog control, with the measurement range configurable via software.

Power switch (hard switch ON/OFF)

10k NTC 3950B external temperature sensor interface

MPPT hardware switch, with an indicator light illuminating upon MPPT activation.

Serial communication TTL level

485 communication AB signal

Can be used simultaneously

Onboard 485 communication interface for more convenient long-distance communication control via 485.

Can also connect our XY-K485X module to achieve 485 bus control

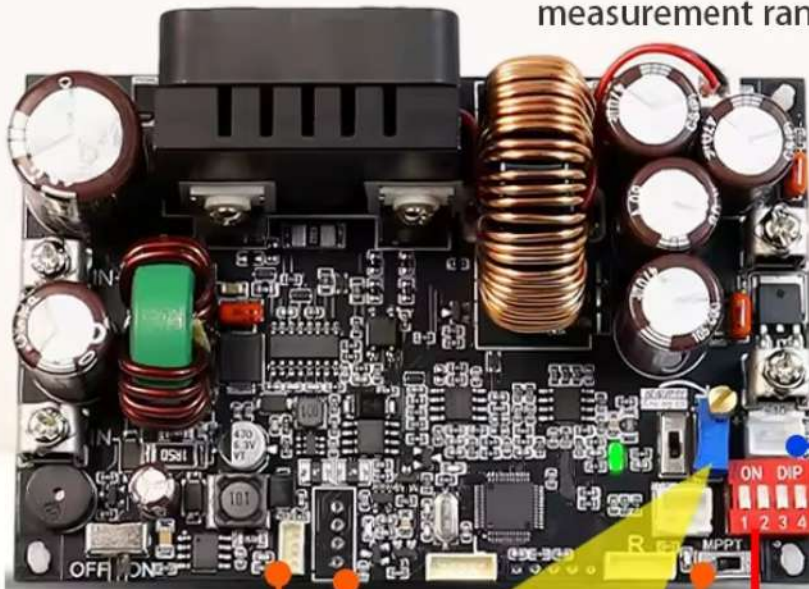
Connect our WIFI module to enable remote control via APP and web-based networking control

Can be connected to our infrared receiver module, with our infrared remote control more efficient and fast

Our 433M wireless remote control module can be accessed, and the remote control can transmit in any direction, which is efficient and fast, and can realize the synchronous control of multiple devices.

- The bare board can be used independently, supporting MPPT and analog 0.3-10V control, with serial communication + on-board 485 communication (both can be used simultaneously).
- The output voltage is adjustable from 0-70V, and the output current is adjustable from 0-25A.

Support 0.3-10V analog control, with the measurement range configurable via software.



Serial communication interface

485 communication interface

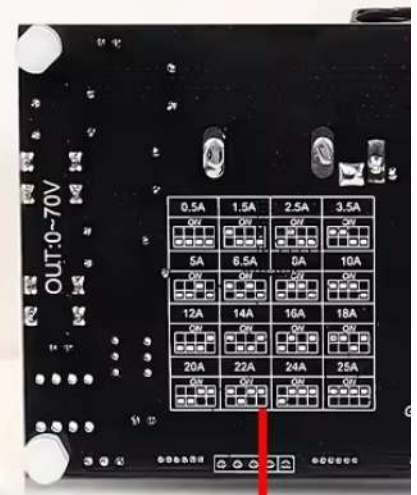


NOTE:

When purchasing a bare device, switches, potentiometers, and dial switches will be welded on. If a set with screen meter is purchased, they are not welded by default.

MPPT ON/OFF

Flip the switch down to light up the green indicator light, entering the potentiometer mode. Conversely, it operates in the digital control meter mode.



0.5A	1.5A	2.5A	3.5A
ON DIP 1 2 3 4	ON DIP 1 2 3 4	ON DIP 1 2 3 4	ON DIP 1 2 3 4
5A	6.5A	8A	10A
ON DIP 1 2 3 4	ON DIP 1 2 3 4	ON DIP 1 2 3 4	ON DIP 1 2 3 4
12A	14A	16A	18A
ON DIP 1 2 3 4	ON DIP 1 2 3 4	ON DIP 1 2 3 4	ON DIP 1 2 3 4
20A	22A	24A	25A
ON DIP 1 2 3 4	ON DIP 1 2 3 4	ON DIP 1 2 3 4	ON DIP 1 2 3 4

In potentiometer mode, the output voltage (constant voltage) can only be controlled through the potentiometer; the constant current value can be set via a dip switch, with a total of 16 gears to meet most scenarios (can also be set via Modbus).

The advantage of setting the constant current value through the dip switch is that it eliminates the need to use a multimeter's current range for measurement, thereby avoiding the potential burnout of the multimeter due to insufficient range of most multimeters' 10A gear.

Product UI function details

When the power supply is on, it is displayed ON



constant voltage CV

Output voltage

Output current

38.90_w power W

13.18^{Ah} capacity Ah

4360_w^h energy Wh

00:06^h time h

043^{°C} temperature °C

On the Main UI, short pressing the Encoder button toggles between displaying output power (W), capacity (Ah), energy (Wh), time (h), temperature (°C), showing them in rotation.

When connected to an external temperature probe (10K 3950B NTC), it will automatically display the probe's temperature.

Constant current display: CC



Constant power display: CW



"OFF" is displayed when the power is off



SET CV 24.00 Set voltage CV

SET CC 06.10^A Set current CC

043^{°C} External probe temperature

When the power is off, the set voltage and current are displayed downward in turn.

Introduction to CV/CC/CW Functions



1 When the constant power function is not turned on, the power supply only has the functions of constant voltage (CV) and constant current (CC), which automatically switch based on the load;

1.1 When the load current is less than the set constant current value, the power supply is in the constant voltage mode, where the output voltage is the set voltage value CV, and the current is adaptive;

1.2 When the load current exceeds the set constant current value, the power supply automatically enters the constant current mode. At this time, the output current is the set constant current value CC, and the voltage is adaptive;

2 After the constant power function is turned on, the constant current value defaults to the maximum value, and the constant voltage value CV serves as the initial voltage (set to a reasonable value based on actual conditions). After the power supply is turned on, the equivalent resistance R of the load is calculated using Ohm's law $R=U/I$. Then, the corresponding voltage can be calculated based on the set constant power value using the power formula $P=U^2/R$. At this point, the constant power point algorithm automatically follows, and the constant power is achieved.

Constant Power Switch and Constant Power Value Setting

1. Press and hold SW on the main UI to enter system set.
2. Press ISET/VSET briefly to switch to the "-CP-" constant power switch option. Press ON to enable constant power and OFF to disable it.
3. After enabling constant power, press ISET briefly on the main interface to modify the value of constant power.

Set UI Parameter Settings within Data Groups

Press and hold the ISET button on the main interface to enter the settings menu.

The first parameter is to select the data group Cd0-9.

Press VSET briefly to select the next parameter, and press ISET briefly to select the previous parameter. After completing the settings, press and hold the ISET/SW/encoder button to exit the settings interface.

...

Select the data group to be set Cd0-Cd9 (default is Cd0).

Set the voltage CV.

Set the current CC

Set the input undervoltage protection value (LVP)

Set the output overvoltage protection value (OVP)

Set the output overcurrent protection value (OCP)

Set the output overpower protection value (OPP)

Set the overcapacity protection value (OAH) default off.

Set the overenergy protection value (OPH) default off

Set the overtime protection value (OHP) default off

Set the overtemperature protection value (OTP) default 95°C

Set the external overtemperature protection value (ETP) default off

Automatically enabled during power-on(POn) default off

Set UI System Parameter Settings

Press and hold the SW button on the main interface to enter the settings menu.

The first parameter is the buzzer setting (bEP). Press VSET briefly to select the next parameter and press ISET briefly to select the previous parameter.

After completing the settings, press and hold the SW/encoder button to exit the settings.

 Turn on/off the buzzer (bEP), default is on	 LCD brightness adjustment (b-L), default is the brightest	 Celsius (°C) and Fahrenheit (°F), selection (C-F), default is °C
 Quick voltage or current adjustment (FET), default is voltage CV	 Set the device address (Add)	 Set the device communication baud rate (bRE), default: 115200
 MPPT solar charging settings (PPT), default is off	 Constant Power CW ON/OFF, Default Off	 Set the charging cut-off current (bTF), default is off
 Calibrate the output voltage (CLU)	 Calibrate the output current (CLA)	 Current zero calibration (ZERO)
 Force the power output to turn off when retrieving data groups (CLOF)	 Restore factory settings (RET)	 Firmware version number (UER)

Data Group Parameter Settings

Press and hold the ISET button on the main interface to enter the settings menu.
The first parameter is to select the data group Cd0-9.

Press VSET briefly to select the next parameter, and press ISET briefly to select the previous parameter. After completing the settings, press and hold the ISET/SW/encoder button to exit the settings interface.

Data Group Selection Cd0-Cd9



Rotate the encoder to select the desired data group Cd0-Cd9.

For example, if Cd2 is selected, subsequent parameter settings will be for the Cd2 data group.

Setting Voltage CV



Briefly press SW or the encoder button to select the position, and rotate the encoder to adjust the value.

For instance, setting 12.00 will result in a CV voltage of 12.00V when this data group is accessed.

Setting Current CC



Briefly press SW or the encoder button to select the position, and rotate the encoder to adjust the value.

For example, setting 6.000 will result in a CC current of 6.000A when this data group is accessed.

Note: The CV and CC modes switch automatically based on the load. When the load reaches the set CC value, it automatically switches to CC mode.

LVP Settings (Input Under-Voltage Protection)



Briefly press SW or the encoder button to select the position, and rotate the encoder to adjust the value.

For instance, if LVP is set to 12.00V, the output will be shut off for protection when the input voltage drops below 12.00V.

After protection, "LUP" will be displayed on the bottom line. Press any button to cancel the alarm. When the input voltage rises above LUP, protection is automatically canceled.

OVP Settings (Over-Voltage Protection)



Short press SW or encoder button to select the position, rotate the encoder to adjust the value.

For instance, if OVP is set to 24.00, when the output voltage exceeds 24.00V, the output will be shut off for protection, thus protecting the load from burnout due to overvoltage.

After protection, "OVP" will be displayed on the bottom line. Press any button to cancel the alarm.

	any button to cancel the alarm.
OCP Setting (Over-Current Protection) 	Short press SW or encoder button to select the position, rotate the encoder to adjust the value. For example, if OCP is set to 2.000, when the output current exceeds 2.000A, the output will be shut off for protection, protecting the load from burnout due to overcurrent. After protection, "OCP" will be displayed on the bottom line. Press any button to cancel the alarm.
OPP Setting (Over-Power Protection) 	Short press SW or encoder button to select the position, rotate the encoder to adjust the value. For instance, if OPP is set to 100.0W, when the output power exceeds 100.0W, the output will be shut off for protection, preventing the load from burnout due to overpower. After protection, "OPP" will be displayed on the bottom line. Press any button to cancel the alarm.
OAH Setting (Over-Capacity Protection) 	Short press the power button to turn on/off the over-capacity protection function. ---- This function is off by default. Short press the power button to activate this function, then short press SW or encoder button to select the position, rotate the encoder to adjust the value. Long press the power button to switch the decimal point position (0.000Ah, 00.00Ah, 000.0Ah, 0000Ah). The maximum setting is 9999Ah. For example, if OAH is set to 2.000Ah, when the cumulative output capacity exceeds 2.000Ah, the output will be shut off for protection. After protection, "OAH" will be displayed on the bottom line. Press any button to cancel the alarm and reset the cumulative capacity.
OPH Setting (Over-Energy Protection) 	Short press the power button to turn on/off the over-energy protection function. ---- This function is off by default. Short press the power button to activate this function, then short press SW or encoder button to select the position, rotate the encoder to adjust the value. Long press the power button to switch the decimal point position (0.000Wh, 00.00Wh, 000.0Wh, 0000Wh, 0.0.0.0Wh (representing 0000KWh)). The maximum setting is 4200KWh. For instance, if OPH is set to 500.0Wh, when the cumulative output energy exceeds 500.0Wh, the output will be shut off for protection. After protection, "OPH" will be displayed on the bottom line. Press any button to cancel the alarm and reset the cumulative energy.

OHP Setting (Over-Time Protection)



Short press the power button to turn on/off the over-time protection function. ---- This function is off by default.

Short press the power button to activate this function, then short press SW or encoder button to select the position, rotate the encoder to adjust the value. The minimum unit is 1 minute, and the maximum setting is 99:59 (99 hours 59 minutes).

For example, if OHP is set to 02:30, when the output is on for more than 2 hours and 30 minutes, the output will be shut off for protection.

After protection, "OHP" will be displayed on the bottom line. Press any button to cancel the alarm and reset the cumulative time.

OTP Setting (Over-Temperature Protection)



Short press SW or encoder button to select the position, rotate the encoder to adjust the value in units of °C or °F (Switch between °C or °F in the system settings interface).

For instance, if OTP is set to 90°C, when the PCB temperature near the power transistor reaches 90°C, the output will be shut off for protection.

After protection, "OTP" will be displayed on the bottom line. Press any button to cancel the alarm. When the temperature drops below OTP, the protection will be automatically canceled.

ETP Settings (External Over-Temperature Protection)



Press the power button briefly to enable/disable the external over-temperature protection function, which is defaulted to "off" when disabled.

To enable this function, briefly press the power button, then press the SW or encoder button to select the position, and rotate the encoder to adjust the value.

For example, if the ETP is set to 60°C, upon connecting an external temperature probe (10K, NTC probe), the output will automatically shut off for protection when the temperature exceeds 60°C.

After protection is activated, "ETP" will be displayed on the bottom row. Press any button to cancel the alarm, and the protection will be automatically lifted when the temperature falls below the ETP setting.

Application Scenario: Attach the external temperature probe to the load (such as a rechargeable battery). When the load temperature exceeds the set temperature, the output will be shut off for protection, effectively preventing the load from overheating and damaging.

PON Settings (Power-On Output)



Rotate the encoder to select between OFF (output off upon power-on) and ON (output on upon power-on).

For example, if PON is set to OFF, then the output is off when the power is just powered on, you need to press the power button to open the output; Otherwise, the output is directly turned on after the power-on.

System Parameter Settings

Press and hold the SW button on the main interface to enter the settings menu.

The first parameter is the buzzer setting (bEP). Press VSET briefly to select the next parameter and press ISET briefly to select the previous parameter.

After completing the settings, press and hold the SW/encoder button to exit the settings.

bEP Settings (Beeper Enable)



Rotary encoder to turn on/off the beeper.

For example, selecting OFF will disable the beeper, and there will be no keystroke prompt tone or alarm one.

b-L Settings (LCD Brightness Adjustment)



Rotary encoder to adjust the brightness level.

1-5 levels, factory default is level 5 (brightest).

C-F Settings (Choice between Celsius °C and Fahrenheit °F)



Rotary encoder to select 'C' or '°F'

Meeting the needs of different countries and regions worldwide.

FET Settings (Quick Adjustment of Voltage, Current, or Power)



Rotary encoder to select CV/CC/OFF/CP.

CV: Quickly adjust voltage using the rotary encoder on the main interface.
CC: Quickly adjust current using the rotary encoder on the main interface.
OFF: No action when rotating the encoder on the main interface.
CP: Quickly adjust power using the rotary encoder on the main interface (when constant power is enabled).

Add Settings (Power Supply Address)



Pressing the SW/encoder button briefly to select the bit, and rotate the encoder to adjust the numerical value.

Value range: 1~247, factory default is 001.

The product with serial communication is a low-cost communication power supply that supports the standard ModBus protocol and can be networked through 485 modules or WIFI modules.

bRE Settings (Communication Baud Rate Settings)



Rotary encoder to set different values from 0-8.

0:9600 1:14400 2:19200
3:38400 4:56000 5:576000
6:115200 7:2400 8:4800

Default baud rate is 115200 at level 6.

PPT Settings (MPPT Solar Charging Settings)



Rotary encoder to turn on/off the MPPT function. OFF: Disable. ON: Enable.

After enabling, press the SW or encoder button briefly to switch to setting the maximum power point coefficient. Rotate the encoder to adjust the coefficient value between 0.75-0.85, with a default of 0.8.

CW Settings (Constant Power CW ON/OFF)



To enable or disable the constant power CW function using a rotary encoder

After enabling constant power, press ISET briefly on the main interface to modify the value of constant power.

bCH Settings (Battery automatic charging threshold setting.)



Press the power button briefly to
turn this function on or off
----This function is off by default.

When the battery voltage falls below
the threshold, the power output is
automatically activated to
commence charging.

For instance, if the threshold is set at
12.00V, when the battery voltage
drops below 12.00V, the power
output is automatically activated to
start charging the battery.

bTF Setting (Charge Cutoff Current)



Press the power button briefly to turn
this function on or off
----This function is off by default.

After enabling this function, press the SW
or encoder button briefly to select the
digit, and rotate the encoder to adjust the
value in mA.

For example, if set to 10mA, when charging
the battery, if the charging current is less
than 10mA, it will be considered as fully
charged, and the output will be disconnect-
ed to prevent overcharging and damaging
the battery.

CLU Setting (Calibrate Output Voltage)



If the output voltage is inaccurate, you
can calibrate it (do not connect any load
during calibration).

1. Press and hold the power button to start
calibration, and "01" will be displayed on
the first line.
2. Measure the output voltage using a high
-precision multimeter and input the
actual voltage into the third line (SW shift
operation is supported).
3. Press the power button briefly to enter
the second step of calibration, and "02"
will be displayed on the first line.
4. Repeat step 2 and input the actual
voltage into the third line.
5. Press the power button again briefly,
and "03" will be displayed on the first line.
Wait for calibration to complete without any
other operation. If successful, "SUC" will be
displayed; if failed, "ERR" will be displayed.

Note: If calibration fails, you can try again.
If you accidentally enter the calibration state,
you can exit by pressing and holding the
power button, and the parameters will not
be saved after exiting.

CLA Setting (Calibrate Output Current)



If the output current is inaccurate, you can
calibrate it.

1. Connect a multimeter in current mode or an
electronic load (the electronic load needs to
be set to the maximum range of the product)
directly to the output terminal.
2. Press and hold the power button to start
calibration, and "01" will be displayed on the
first line. Input the actual current value from t
he multimeter or electronic load into the third
line (SW shift operation is supported).
3. Press the power button briefly to enter the
second step of calibration, and "02" will be
displayed on the first line.
4. Repeat step 2 and input the actual current
into the third line.
5. Press the power button again briefly and wait
-for calibration to complete without any other
operation. If successful, "SUC" will be displayed;
if failed, "ERR" will be displayed.

Note: If calibration fails, you can try again. If you
accidentally enter the calibration state, you can
exit by pressing and holding the power button,
and the parameters will not be saved after exiting.

ZERO Setting (Current Zero Calibration)



If there is a small current within 100mA when the output is not connected to a load, you can press and hold the power button to calibrate the zero point. Do not connect any load to the output terminal during zero calibration

CLOF Setting (Force Power Output Off When Switching Data Sets)



Rotate the encoder to select ON or OFF for this function.

ON: Enable this function. After enabling, when switching data sets, the power will be forcibly turned off to prevent high voltage from damaging the load.

OFF: Disable this function. After disabling, the power switch state will be maintained when switching data sets.

POFF (ShutDown Function)



ON: Enable the ShutDown function. Press and hold the Power Key for 5 seconds to shut down; In the shutdown state, click the Power Key to start up;

OFF: Disable the shutdown function;

RET Setting (Restore Factory Settings)



Press and hold the power button ---- it stops blinking to restore factory settings.

WIFI Config (Need WF-POW module)



Modify pairing mode with encoder

-----: Invalid Wifi

TOH: Touch Pair

AP: AP Pair

ROU: Connect Router

nET: NetWorking

AnA Setting (Analog Control)



Analog Control Voltage Range, which can be self-configured.

Adjustable analog voltage range from 2 to 10V;

For example, with an analog voltage range of 0~5V and a power output range of 36V, when the signal voltage is 2V, the output voltage = $2V / (5V - 0.3V) * 36V = 15.32V$.

When the signal voltage is below 0.3V, the output voltage is 0V. When the signal voltage exceeds the set maximum value, the output voltage is the set maximum range voltage.

UER View (Firmware Version Number)



The firmware version number of the program.

The product will support firmware upgrades to access new features.

Instructions for Use

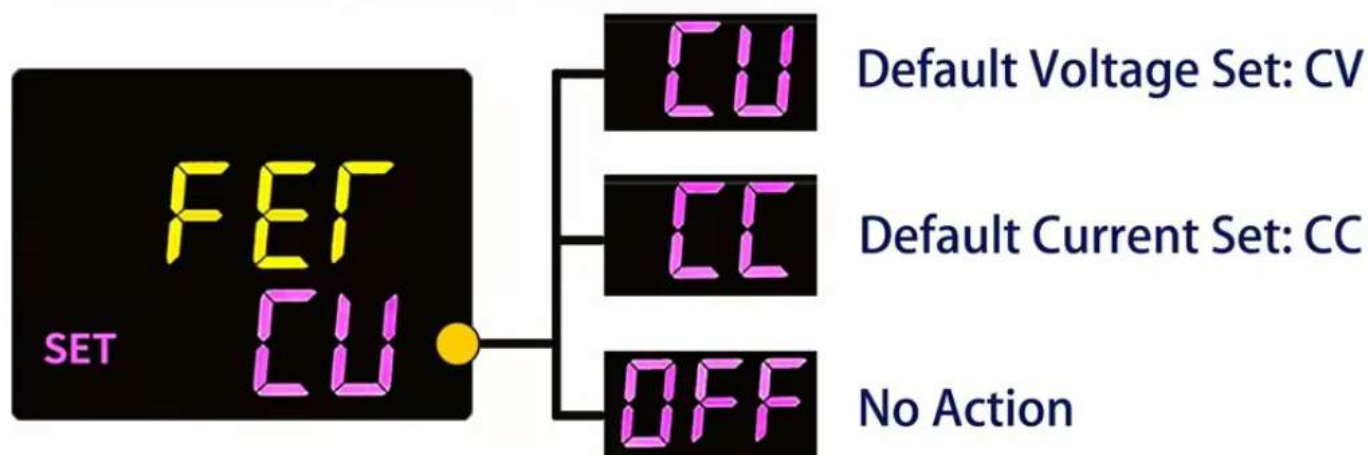


1. Setting Voltage and Current:



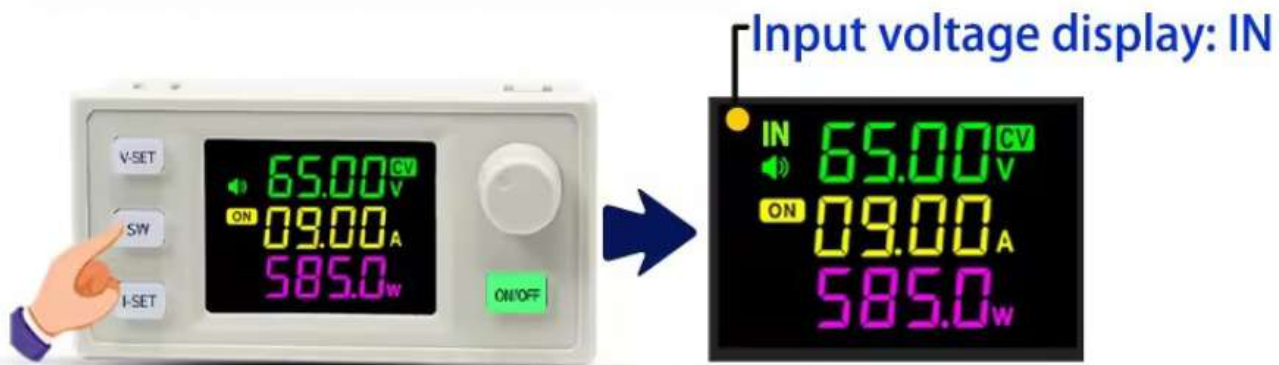
On the main UI, short press the **VSET** button to set the voltage. The LCD will display "SET" in the lower row, and "CV" will flash to indicate that the voltage setting position is selected and blinking. Then, short press the **SW** button or the encoder button to switch the voltage setting position. Adjust the value by rotating the encoder. After setting is complete, short press the **VSET** button to exit and save. To set the current, short press the **ISET** button, and the setting steps are the same as for voltage.

2. Quick Setting of Voltage or Current:



In the system parameter settings UI, set the parameter FET to CV or CC. Rotate the encoder on the main UI to enter the voltage or current setting UI. Rotate the encoder to quickly set the voltage or current.

3. Input/output voltage display:



On the main UI, press the **SW** button briefly to switch between input and output voltage displays.

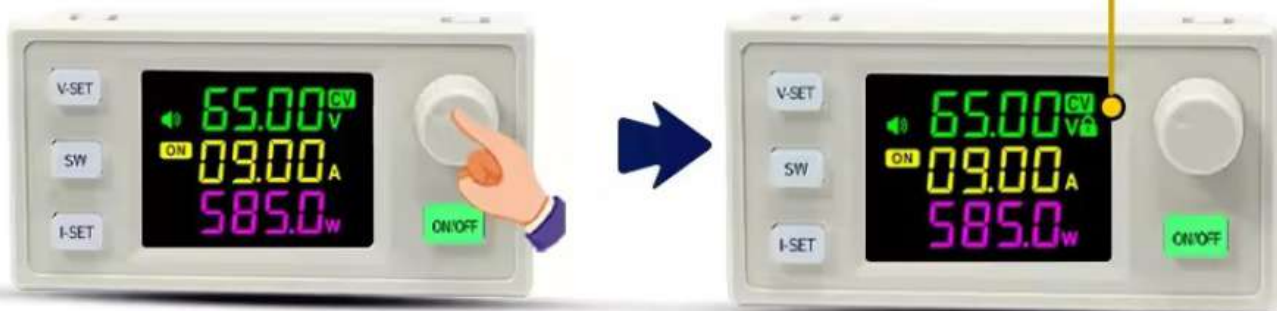
4. Checking power (W)/capacity (Ah)/energy (Wh)/time (h):



On the main UI, press the encoder button briefly to switch between displaying power (W)/capacity (Ah)/energy (Wh)/time (h)/temperature (°C) in rotation.

5. key lock:

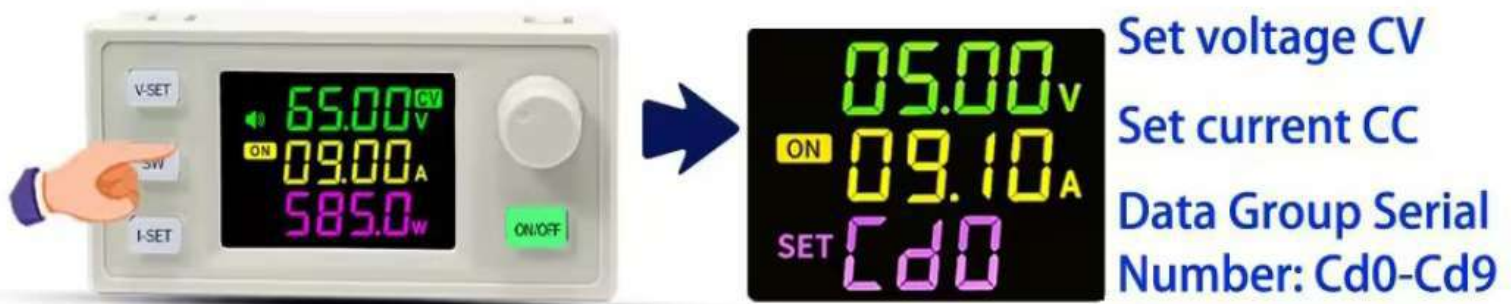
Display lock symbol



On the main UI, press and hold the encoder button for 2 seconds to lock the set voltage and current to prevent misoperation; press and hold the encoder button for 2 seconds after locking to unlock.

6.Data Group Function

This product has a total of 10 data groups from Cd0 to Cd9. Press and hold the VSET button on the main UI to access the data group UI



You can press VSET/ISET briefly to switch between constant voltage (CV) and constant current (CC) settings. It supports quick viewing and modification of voltage and current in the data group, as well as SW shift operation.

After confirming the data group, press and hold the VSET/SW button or press the encoder button briefly to select the desired data group.